UNIVERSITI TEKNOLOGI MARA
FINAL EXAMINATION

COURSE	COMPUTER ORGANIZATION AND ARCHITECTURE
COURSE CODE	CSC429
EXAMINATION	NOVEMBER 2026
TIME	3 HOURS

INSTRUCTIONS TO CANDIDATES

1. This question paper consists of two (2) parts: PART A (10 Questions) PART B (1 Question)
2. Answer ALL questions in the Answer Booklet. Start each answer on a new page.
3. Do not bring any material into the examination room unless permission is given by the invigilator.
4. Please check to make sure that this examination pack consists of:
   i) the Question Paper
   ii) an Answer Booklet - provided by the Faculty
5. Answer ALL questions in English.

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO

This examination paper consists of 5 printed pages.

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PART A (80 MARKS)

Answer ALL questions in this part.

QUESTION A1 **(5 marks)**

a) List TWO (2) advantages of SMP organization.

b) Define cluster organization.

c) Give ONE (1) benefit of cluster organization.

QUESTION A2 **(5 marks)**

Given the following Figure 2.

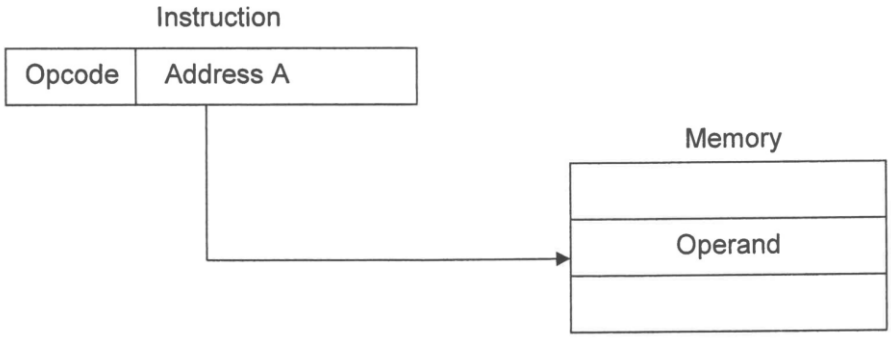


Figure 2

a) Label the addressing technique illustrated in Figure 2.

b) Explain how the addressing technique in 9(a) is used.

c) List ONE (1) advantage and ONE (1) disadvantage of the addressing technique in 9(a).

QUESTION A3 **(5 marks)**

The advancement of technology through different computer generations has focused on improving performance, speed, and efficiency. From vacuum tubes in the first generation to modern AI-powered systems, each generation introduced faster processing, smaller components, and greater reliability.

a) Name the inventor of the following devices:

i. Mechanical Calculating Machine

ii. Analytical Machine iiD IAS Machine

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b) Why are the machines of later computer generation designed for general purposes instead of the eadier computer generation that is more specific in function?

QUESTION A4 **(5 marks)**

Convert -6.75 to IEEE 754 single-precision format. Give the answer in hexadecimal and show the sign, exponent and mantissa fields.

QUESTION A5 **(10 marks)**

Describe the specificfunction ofdifferent types of registers.

QUESTION A6 **(10 marks)**

Keyboard is the most common input/output device to communicate between computer and user.

a) What is the input/output module category for keyboard?

b) State TWO (2) other devices that are in the similar input/output module category?

c) What is the basic unit of exchange for keyboard?

d) Discuss TWO (2) types of characters.

QUESTION A7 **(10 marks)**

Answer both parts. Show your complete workflow.

a) Convert -11.375 to IEEE 754 single precision. Give the answer in hexadecimal.

b) Convert the IEEE 754 single-precision value 0x40500000 to decimal.

QUESTION A8 **(10 marks)**

a) Define interrupt.

b) Describe TWO (2) different types of interrupt.

c) Discuss ONE (1) advantage and ONE (1) disadvantage of interrupt

QUESTION A9 **(10 marks)**

Describe the function of each register in the computer system.

QUESTION A10 **(10 marks)**

Question 7 is based on the following scenarios. While cleaning his house, Ahmad found an old cassette tape of a popular 80's band. He played the tape and listened to several songs according to the listed tracks. Hafiz was tasked to research on the history of Malay culture by his professor. He went to the National Archive to search for information and manage to access one the resources which was stored in a floppy disk. Sara is planning to buy a new computer. She aims to get one with good processing speed. While researching for computer specifications, she found out that larger cache space has a direct effect towards processing performance. Hana loves playing role-playing games (RPG) and is planning to buy a newly released game that requires a high-speed RAM. But, she is devastated when she found out that her game console does not support the minimum RAM requirement of the game.

a) Based on the above scenarios, identify the access method for each storage devices/memory mentioned in (i), (ii), (iii) and (iv).

b) Describe SRAM and DRAM.

c) Which level of cache is the fastest in a three-level cache architecture? Which type of RAM is used to implement the cache?

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PART B (20 MARKS)

Answer the question in this part.

QUESTION B1 **(20 marks)**

Question 1 is based on Figure 2 and Table 1.

$$y = A - \frac{2C}{B + A^2} * D$$

Figure 2

Write an appropriate program using correct assembly language notations to compute the equation shown in Figure 2 using instructions for one-address, two-address, and three- address. The available instructions are provided in Table 1. The register labeled A, B, C, and D are fixed, while registers x, y, and z can be used temporarily. The final answer must be stored in register y.

Table 1 : Machine instruction

One-address	Two-address	Three-address
LOADp	MOV (p,q)	ADD (p.q.r)
STORp	ADD (p,q)	SUB (p,q,r)
ADDp	SUB (p,q)	MUL (p.q.r)
SUBp	MUL (p,q)	DIV(p,q,r)
MULp	DIV(p,q)	
DIVp		